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processing or by intentional damage. The atomically sharp nature of these defects leads to large momentum scattering.

Electrostatic potential fluctuations, created by a random distribution of charges in the substrate by molecules adsorbed on the nanotube or by processing residues, can induce both short- and long-range disorder. Mechanical deformations include local strains and twists, which affect the local band gap as well as kinks that act as conductance barriers.

CNTs possess two kinds of inductances, the magnetic (or mutual) inductance and the kinetic (or self) inductance. Kinetic inductance dominates mutual inductance in a 1D structure like CNTs and hence we only consider kinetic inductance in the model. CNTs have two modes of propagation, with two electrons in each mode (spin up and spin down).

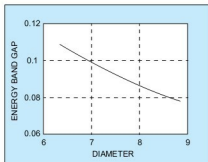


Fig. 3: Variation of band gap with diameter

Larger diameters have a higher leakage and are harder to switch off. Smaller diameters have a 5x decrease in speed than larger-diameter CNTs.

The reason for the usage of atomic force microscopy (AFM), transmission electron microscopy (TEM) and scanning electron microscopy (SEM) in nanotubes analysis is very simple. AFM is a powerful tool in manipulating and characterising the topography of nanostructures. TEM has been traditionally applied to characterisation of the interior structures of nanomaterials. Scanning electron microscope with high resolution is a powerful instrument for imaging of fine structures and morphology at high magnification.

To sum up

The integration of CNTs can be achieved in micro/nano electronic devices if we manage to fabricate nanotubes of particular type, length and diameter in optimised manner. The interdisciplinary science of CNT can be well understood for application in electronics. Though the process is still at embryonic stage, in the near future we will achieve the goal of fruitful integration in electronic devices. ●

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